

Generation Interconnection System Impact Study Report

For Queue Project Q18-1088

Will County 100 kV
December 31, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1088. The Interconnection Customer proposes a 42.8 MW battery energy storage facility interconnecting at the Will County 100 kV substation in the Illinois area. The request was received on October 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 269 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$257,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Golden Gate – Ege 100kV	Dekalb – Jones Chapel 500kV	109.6	340	5.3
Ege – Cuba City 100kV L	Jones Chapel – Dekalb 500kV	108.6	355	5.3
Farmersville – Kendallville 100kV L	Jones Chapel Substation – Dekalb 500kV line	105.8	409	5.3
Royce – Spring Brook 100kV	Brown Plant Substation 100-1 kV Autotransformer	103.3	135	32.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ege – Golden Gate 100kV	Reconductor line	\$1,697,000	0% (\$5,000)
Ege – Cuba City 100kV L	Reconductor line	\$2,216,000	0% (\$7,000)
Farmersville – Kendallville 100kV	Reconductor line	\$2,805,000	0% (\$9,000)
Royce – Spring Brook 100kV	Reconductor line	\$504,000	47% (\$236,000)
	Total allocated to project		\$257,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.